DASAR OPERASI FILTERING

Anny Yuniarti - 2026

AGENDA

- Image Formation
- Filtering Basics

IMAGE FORMATION

Pengolahan Citra Digital

CITRA DIGITAL



$F(x, y)$

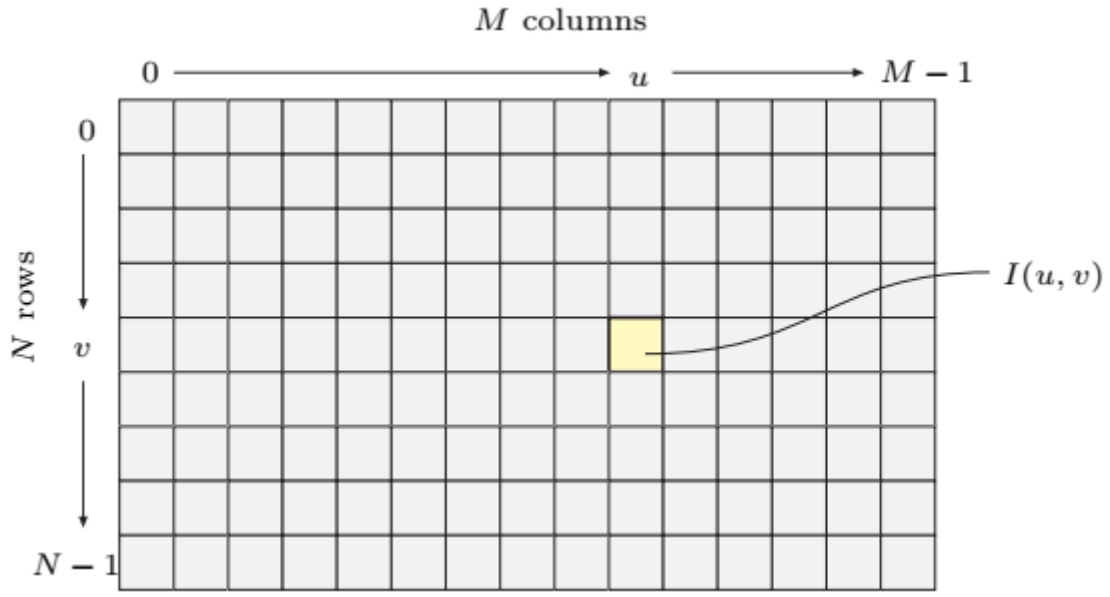
$\rightarrow$

148	123	52	107	123	162	172	123	64	89 $\cdots$
147	130	92	95	98	130	171	155	169	163 $\cdots$
141	118	121	148	117	107	144	137	136	134 $\cdots$
82	106	93	172	149	131	138	114	113	129 $\cdots$
57	101	72	54	109	111	104	135	106	125 $\cdots$
138	135	114	82	121	110	34	76	101	111 $\cdots$
138	102	128	159	168	147	116	129	124	117 $\cdots$
113	89	89	109	106	126	114	150	164	145 $\cdots$
120	121	123	87	85	70	119	64	79	127 $\cdots$
145	141	143	134	111	124	117	113	64	112 $\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

$I(u, v)$



KOORDINAT CITRA DIGITAL



RESOLUSI CITRA (SPASIAL DAN INTENSITAS)

- Resolusi spasial direpresentasikan dengan banyaknya titik (piksel) pada satu satuan jarak, misalnya *dots per inch* (dpi)
  - dpi digunakan di industri percetakan di US
    - Koran dicetak dengan resolusi 75 dpi
    - Majalah: 133 dpi
    - Brosur: 175 dpi
    - Buku: 2400 dpi

RESOLUSI SPASIAL

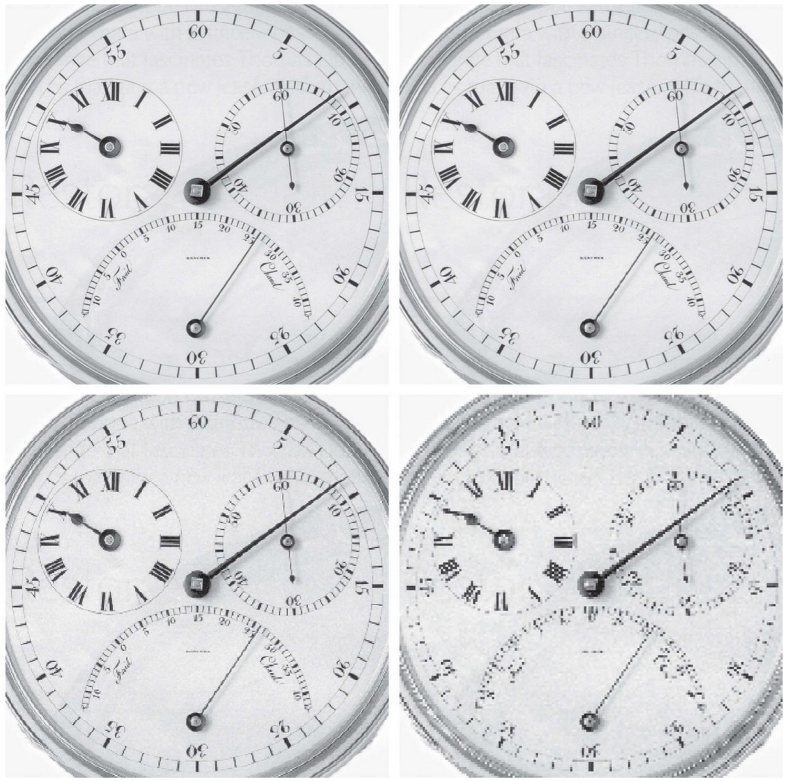
(a) 930 dpi

(b) 300 dpi

(c) 150 dpi

(d) 72 dpi

a b
c d



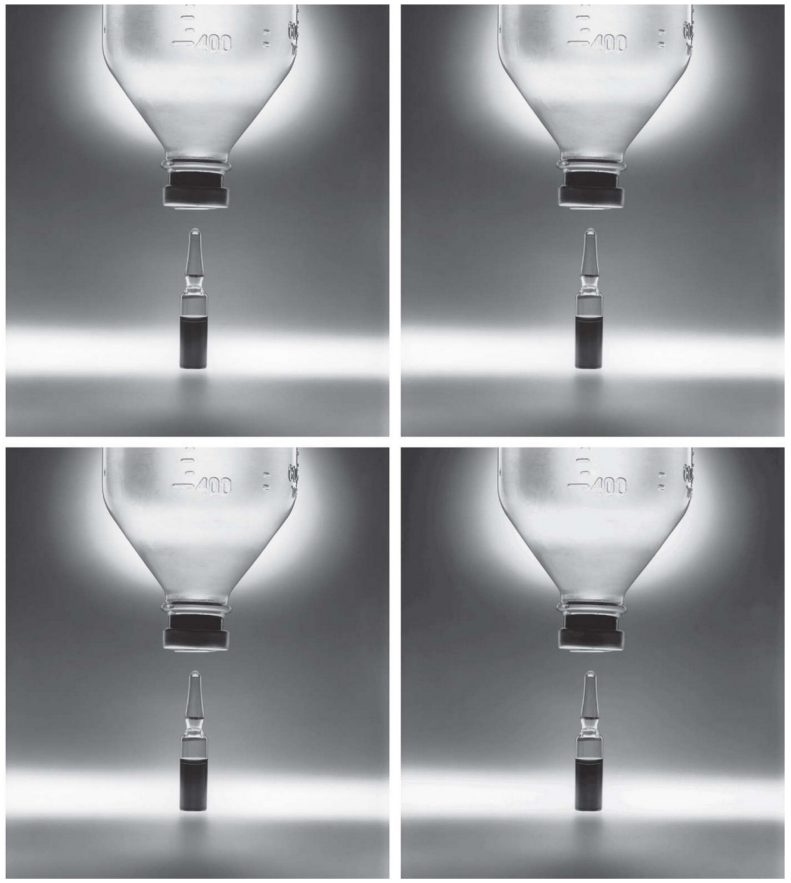
RESOLUSI INTENSITAS

Karena pertimbangan hardware, level intensitas umumnya bilangan pangkat 2, misalnya 8 bits

Resolusi intensitas 8 bits = 256 level intensitas

RESOLUSI INTENSITAS

a b
c d



(a) 256-level image

(b) 128-level

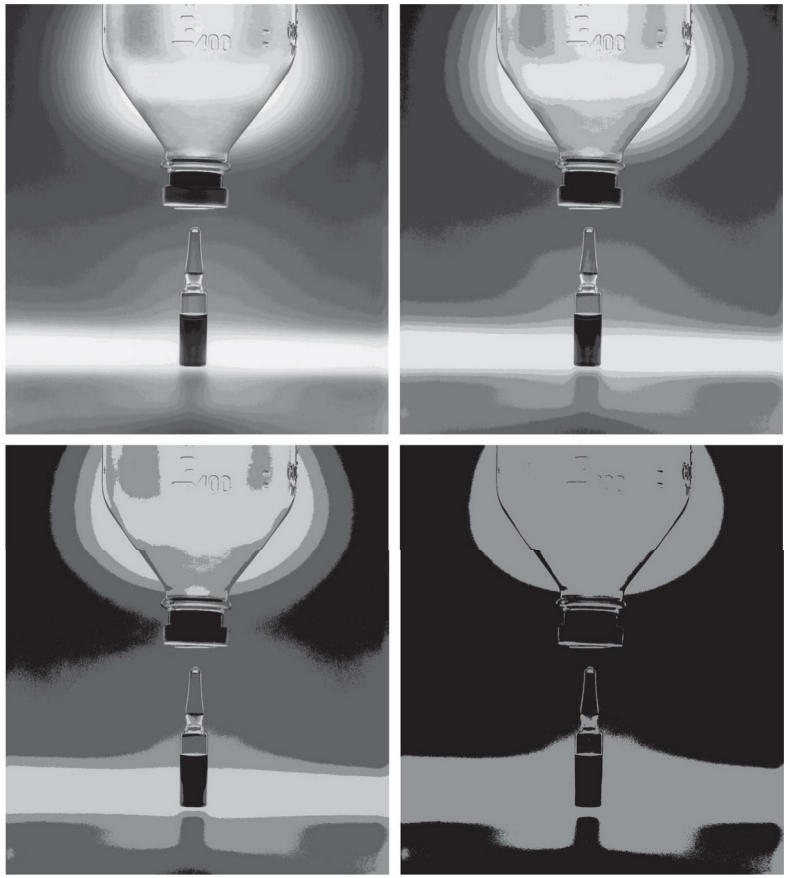
(c) 64-level

(d) 32-level

RESOLUSI INTENSITAS

e. 16-level image
f. 8-level
g. 4-level
h. 2-level

e f
g h



BIT DEPTHS OF COMMON IMAGE TYPES AND TYPICAL APPLICATION DOMAINS

Grayscale (Intensity Images):

Chan.	*Bits/Pix.*	*Range*	*Use*
1	1	$[0, 1]$	Binary image: document, illustration, fax
1	8	$[0, 255]$	Universal: photo, scan, print
1	12	$[0, 4095]$	High quality: photo, scan, print
1	14	$[0, 16383]$	Professional: photo, scan, print
1	16	$[0, 65535]$	Highest quality: medicine, astronomy

Color Images:

Chan.	*Bits/Pix.*	*Range*	*Use*
3	24	$[0, 255]^3$	RGB, universal: photo, scan, print
3	36	$[0, 4095]^3$	RGB, high quality: photo, scan, print
3	42	$[0, 16383]^3$	RGB, professional: photo, scan, print
4	32	$[0, 255]^4$	CMYK, digital prepress

Special Images:

Chan.	*Bits/Pix.*	*Range*	*Use*
1	16	$[-32768, 32767]$	Integer values pos./neg., increased range
1	32	$\pm 3.4 \cdot 10^{38}$	Floating-point values: medicine, astronomy
1	64	$\pm 1.8 \cdot 10^{308}$	Floating-point values: internal processing

DASAR OPERASI FILTERING

Domain Spasial

PRAPROSES CITRA DIGITAL

- Perbaikan kontras dengan fungsi transformasi intensitas
- Filter spasial lowpass dan highpass
- Metode kombinasi filter spasial

PRAPROSES CITRA DIGITAL

Perbaikan kontras dengan fungsi transformasi intensitas

PERMASALAHAN

Low contrast image



TUJUAN

Better contrast image



TUJUAN

Better contrast image



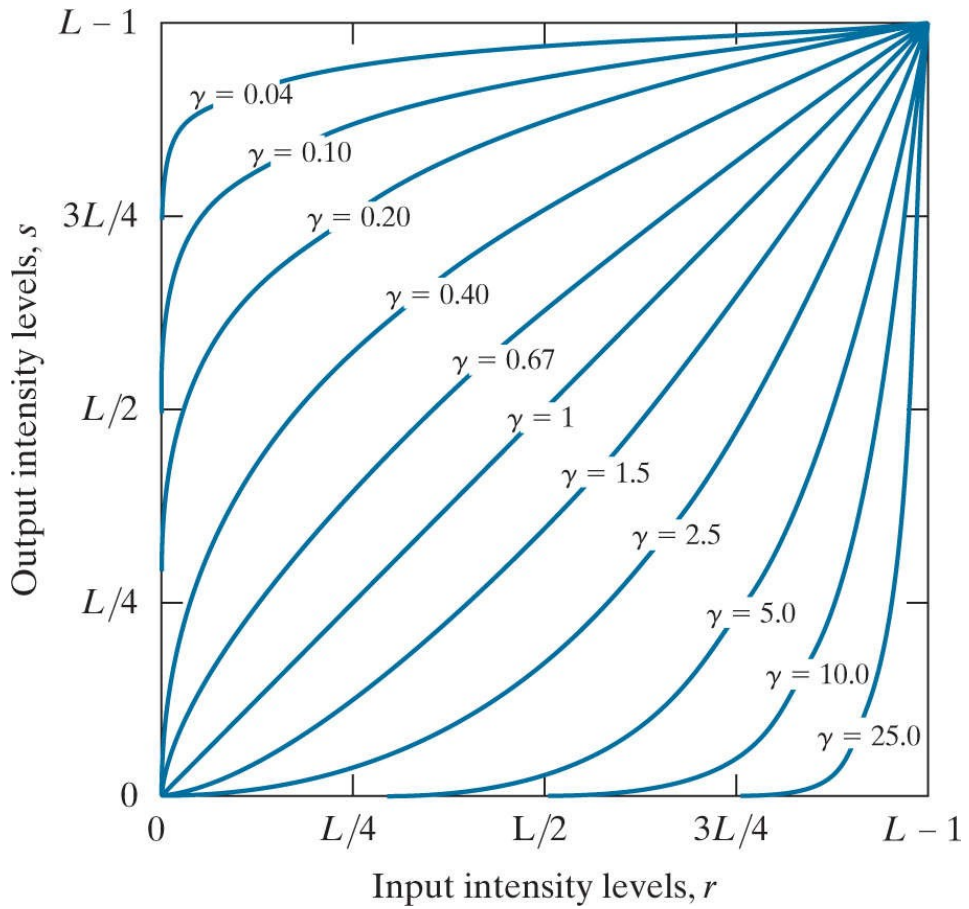
TUJUAN

Better contrast image



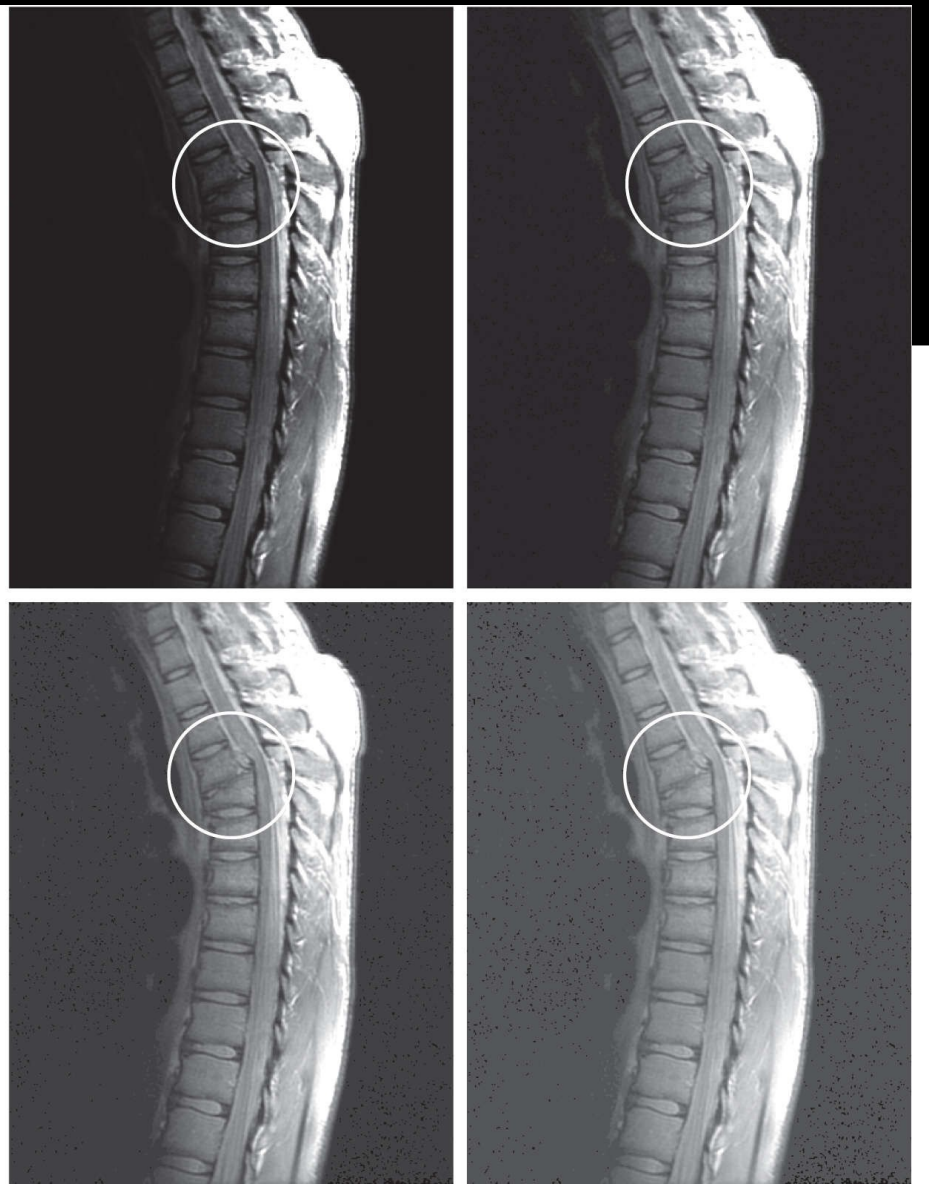
METODE

Transformasi Power-Law (Gamma): mengubah intensitas citra input dengan rumus $s = cr^{\gamma}$



CONTOH

a b
c d



(a) MRI tulang dengan area fraktur yang dilingkari

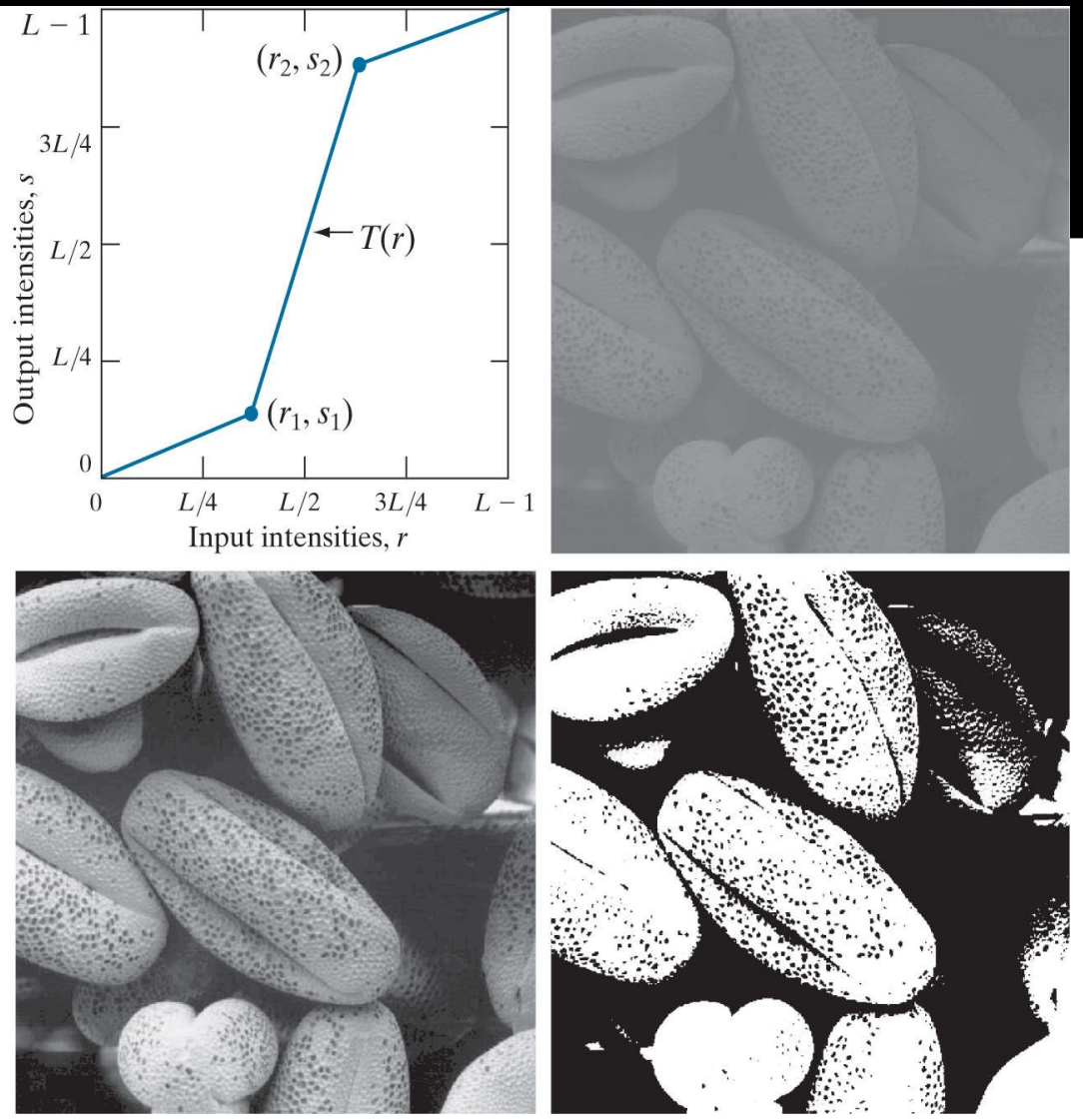
(b) Hasil transformasi gamma dengan nilai gamma (γ) = 0,6

(c) Hasil transformasi gamma dengan nilai gamma (γ) = 0,4

(d) Hasil transformasi gamma dengan nilai gamma (γ) = 0,3

CONTRAST STRETCHIN

a b
c d



(a) Piecewise linear transformation function.

(b) A low-contrast electron microscope image of pollen, magnified 700 times.

(c) Result of contrast stretching.

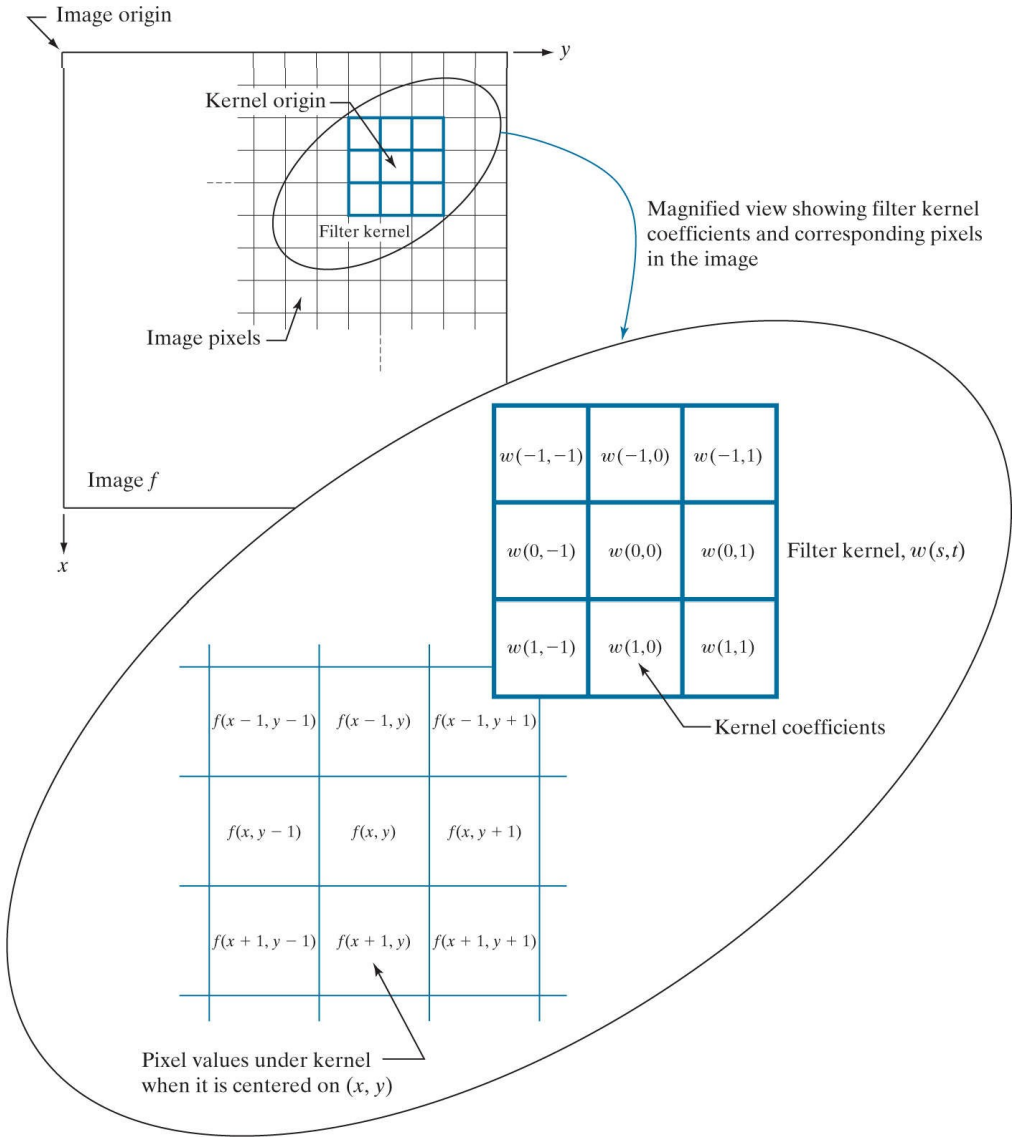
(d) Result of thresholding.

PRAPROSES CITRA DIGITAL

Filter spasial lowpass dan highpass

LINEAR SPATIAL FILTERING USING 3X3 KERNEL

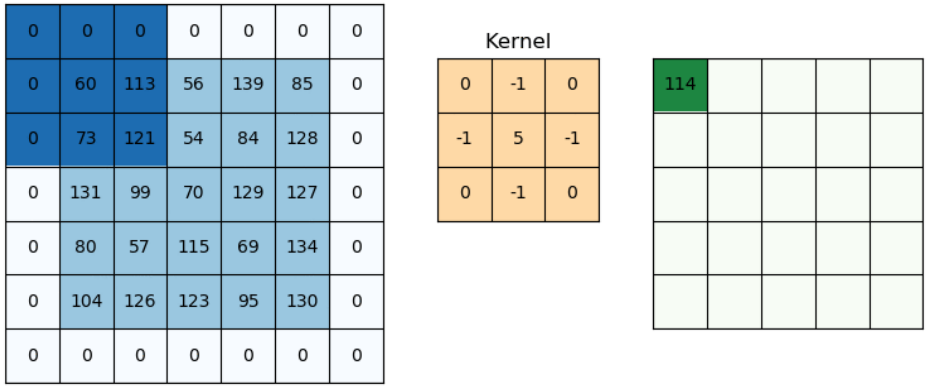
As coordinates *x* and *y* are varied, the center of the kernel moves from pixel to pixel, generating the filtered image, *g*, in the process



$$g(x, y) = w(-1, -1)f(x-1, y-1) + w(-1, 0)f(x-1, y) + \ldots + w(0, 0)f(x, y) + \ldots + w(1, 1)f(x+1, y+1)$$

LINEAR SPATIAL FILTERING USING 3X3 KERNEL

As coordinates *x* and *y* are varied, the center of the kernel moves from pixel to pixel, generating the filtered image, *g*, in the process



$$g(x, y) = w(-1, -1)f(x-1, y-1) + w(-1, 0)f(x-1, y) + \ldots + w(0, 0)f(x, y) + \ldots + w(1, 1)f(x+1, y+1)$$

SMOOTHING (AVERAGING) KERNELS (FILTERS)

- used to reduce sharp transitions in intensity
- can be used for noise reduction

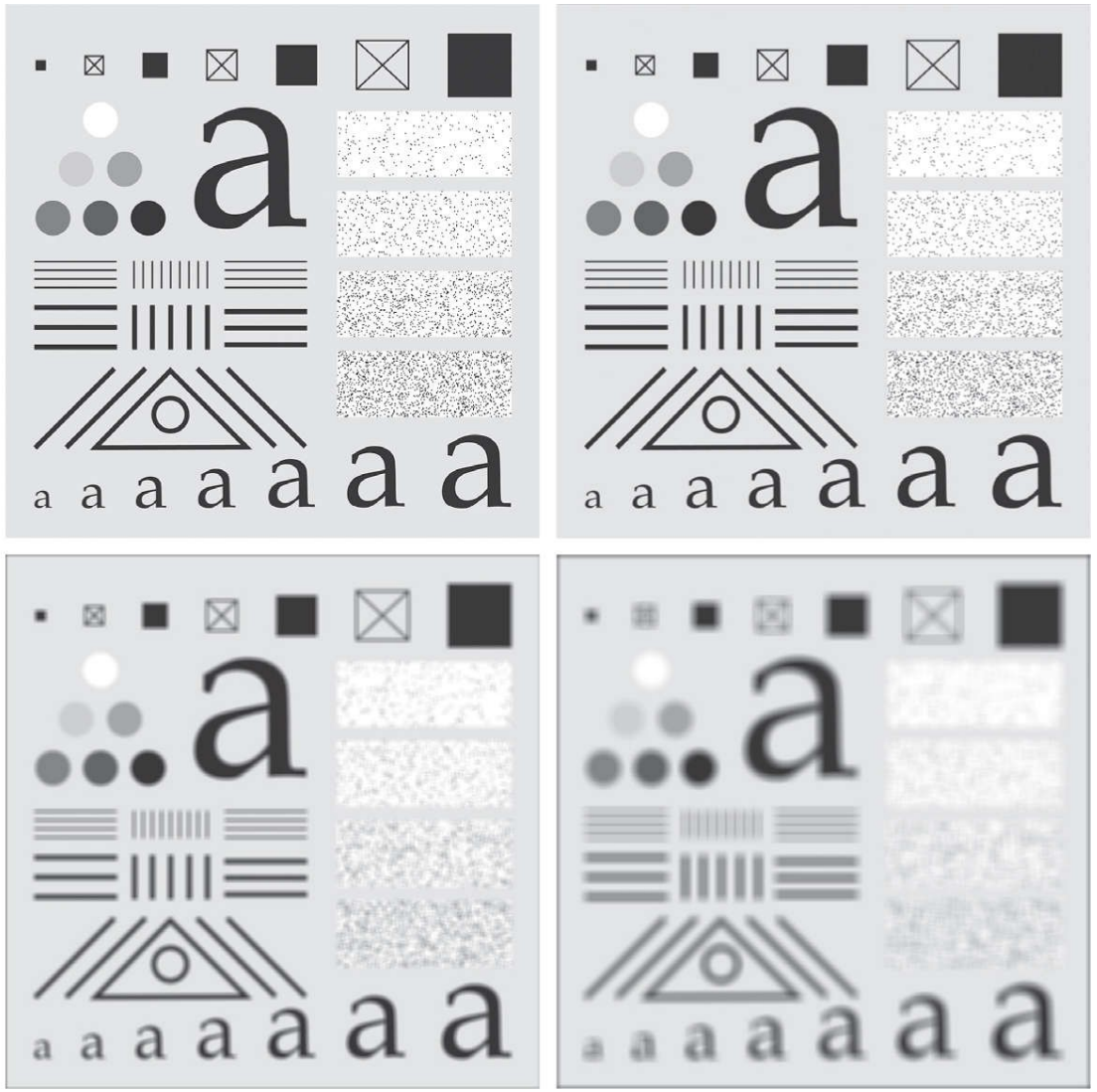
a b

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad \frac{1}{4.8976} \times \begin{bmatrix} 0.3679 & 0.6065 & 0.3679 \\ 0.6065 & 1.0000 & 0.6065 \\ 0.3679 & 0.6065 & 0.3679 \end{bmatrix}$$

(a) is a *box* kernel; (b) is a *Gaussian* kernel.

a b
c d

(a) Test pattern of size pixels.
(b)-(d) Results of lowpass filtering with **box kernels** of sizes and respectively

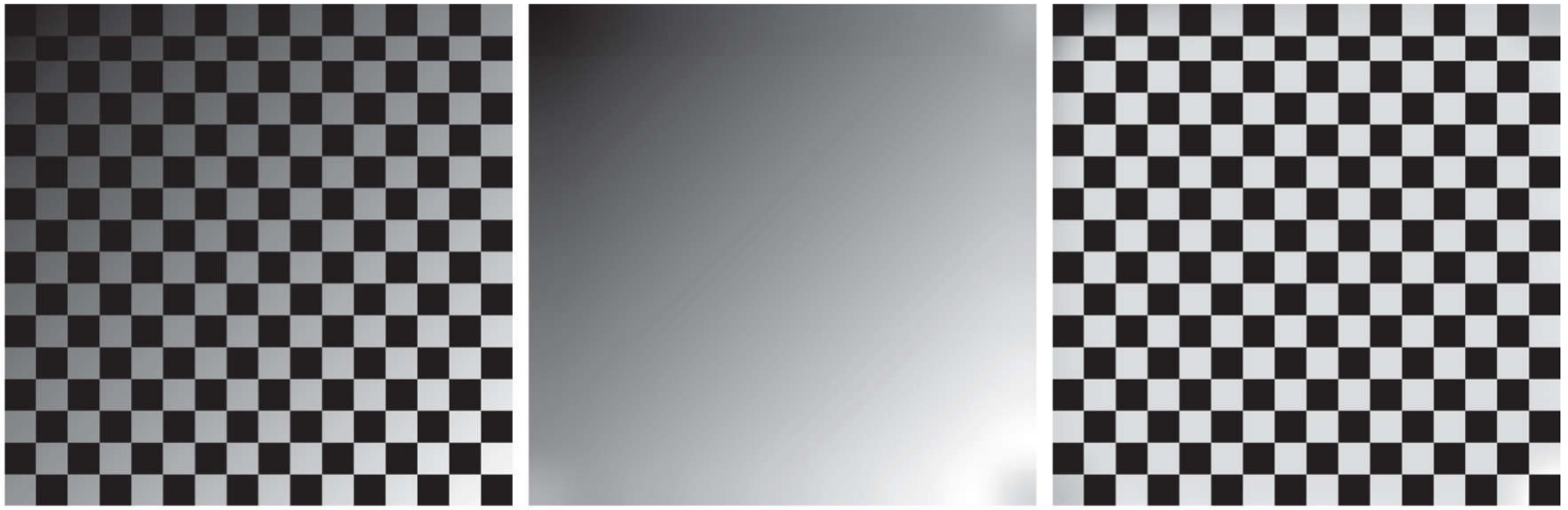




a b c

(a)A test pattern of size 1024 x 1024.
(b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21 x 21, with standard deviations = 3.5
(c) Result of using a kernel of size 43 x43, with standard deviations = 7.

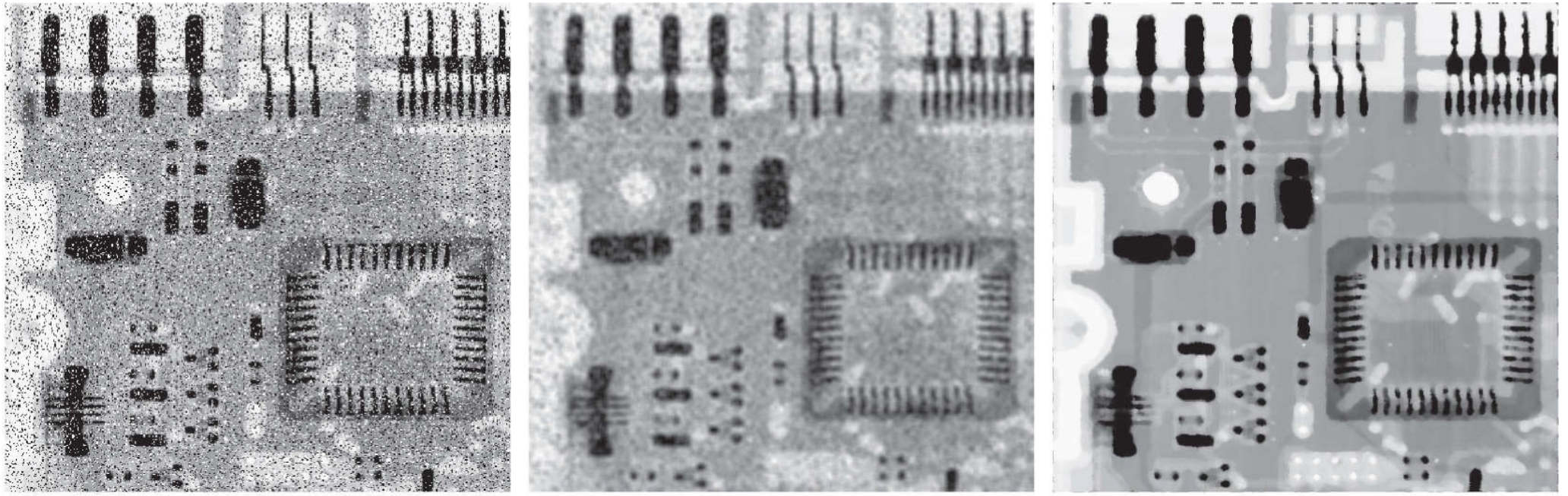
SHADING CORRECTION



a b c

(a) Image shaded by a shading pattern oriented in the direction. The image size: 1024 x 1024. The inner square sizes: 128 x 128.
(b) Estimate of the shading patterns obtained using lowpass filtering. The result of filtering the image with a 512 x 512 Gaussian kernel with standard deviations = 128.
(c) Result of dividing (a) by (b).

MEDIAN FILTERS



a b c

(a) X-ray image of a circuit board, corrupted by salt-and-pepper noise.
(b) Noise reduction using a 19 x 19 Gaussian lowpass filter kernel with standard deviations = 3.
(c) Noise reduction using a 7 x 7 median filter.

SHARPENING USING THE SECOND DERIVATIVE – THE LAPLACIAN

$$\nabla^2 f(x,y) = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)$$

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

a b c d

(a) Laplacian kernel.
(b) Kernel used to implement an extension of this equation that includes the diagonal terms.
(c) and (d) Two other Laplacian kernel

CONTOH

a b
c d

(a) Blurred image of the North Pole of the moon.

(b) Laplacian image obtained using the Laplacian kernel.

(c) Image sharpened using the following equation with c = -1

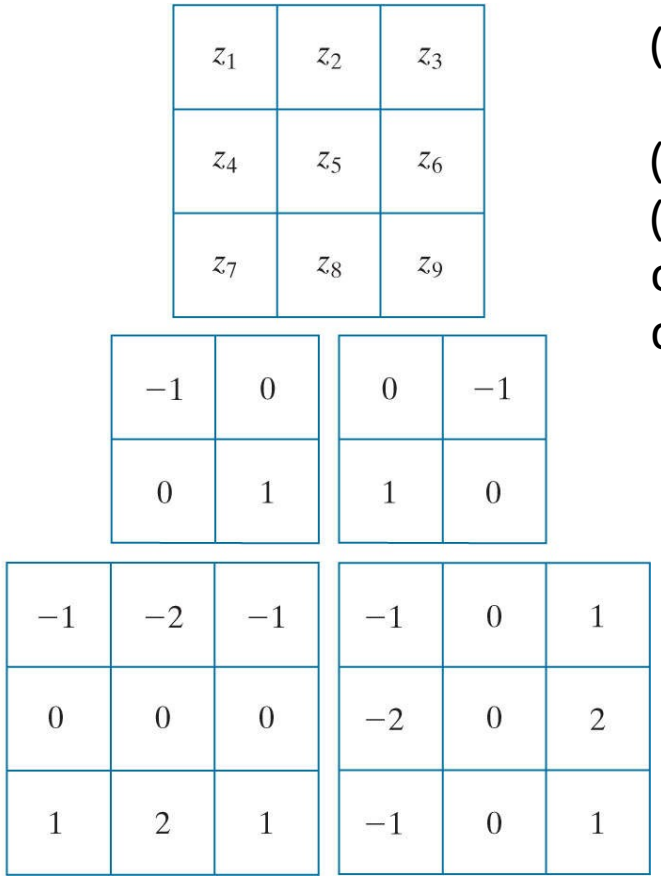
$$g(x,y) = f(x,y) + c[\nabla^2 f(x,y)]$$

(d) Image sharpened using the same procedure, but with the kernel that includes the diagonal terms.



SHARPENING USING THE FIRST DERIVATIVE – THE GRADIENT

a
b c
d e



(a) A region of an image, where the *z*s are intensity values.
(b) – (c) Roberts cross-gradient operators.
(d)–(e) Sobel operators. All the kernel coefficients sum to zero, as expected of a derivative operator.

$$g_x = \frac{\partial f}{\partial x} = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$g_y = \frac{\partial f}{\partial y} = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

SHARPENING USING THE FIRST DERIVATIVE – THE GRADIENT

The Image Gradient Magnitude

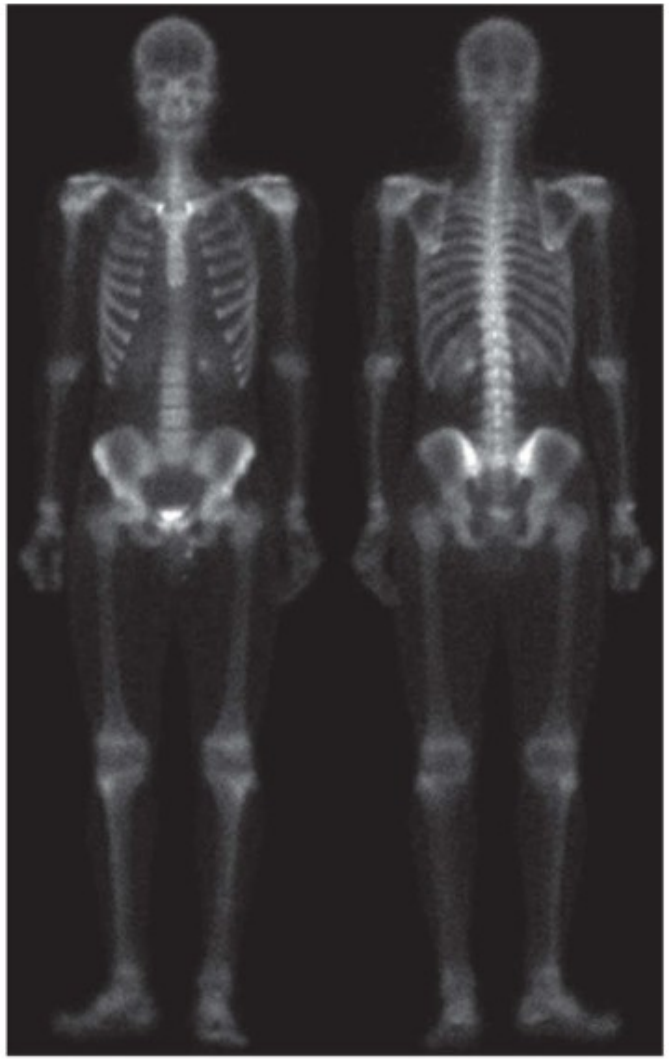
$$M(x,y) = [g_x^2 + g_y^2]^{\frac{1}{2}} = [[(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)]^2 + [(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)]^2]^{\frac{1}{2}}$$

PRAPROSES CITRA DIGITAL

Metode kombinasi filter spasial

IMAGE OF WHOLE BODY BONE SCAN

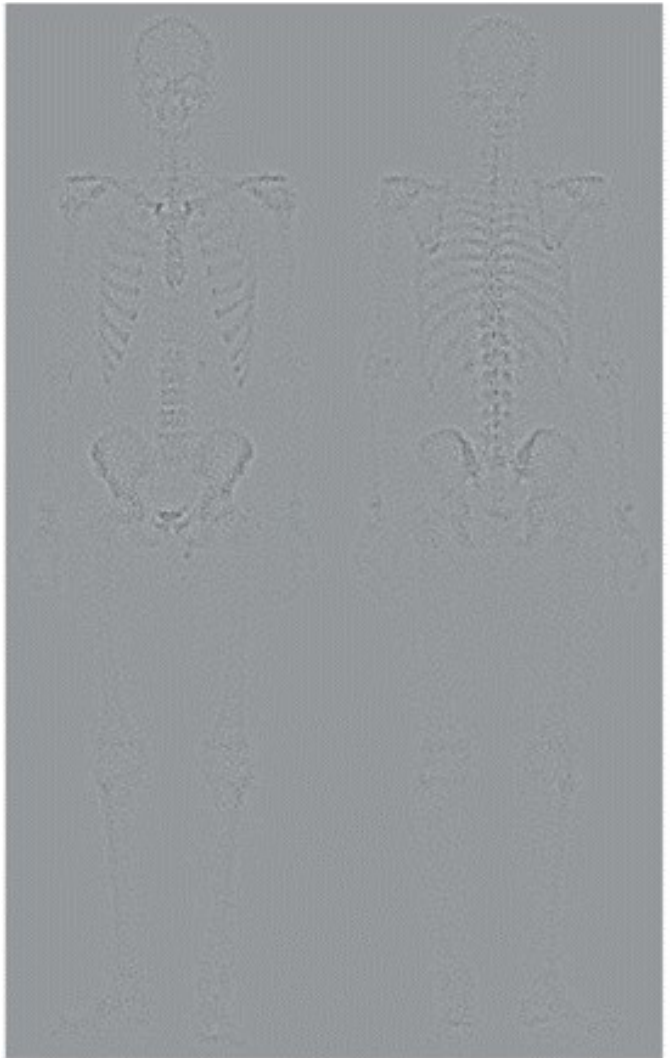
used to detect diseases such as bone infections and tumors



THE LAPLACIAN OF THE IMAGE OF WHOLE BODY BONE SCAN

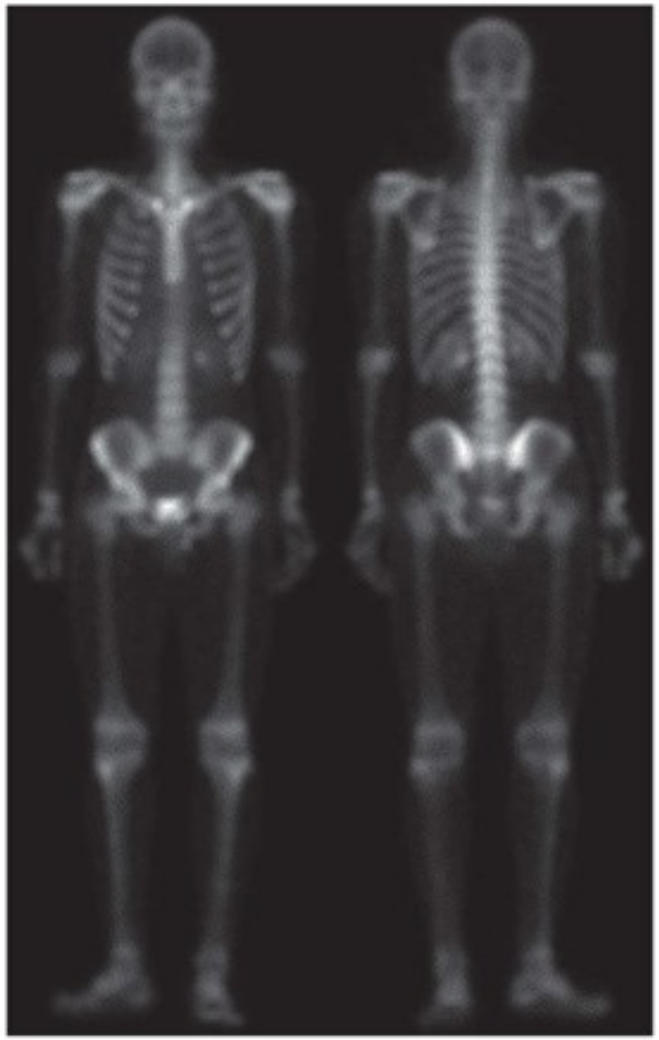
obtained using the following kernel :

−1	−1	−1
−1	8	−1
−1	−1	−1

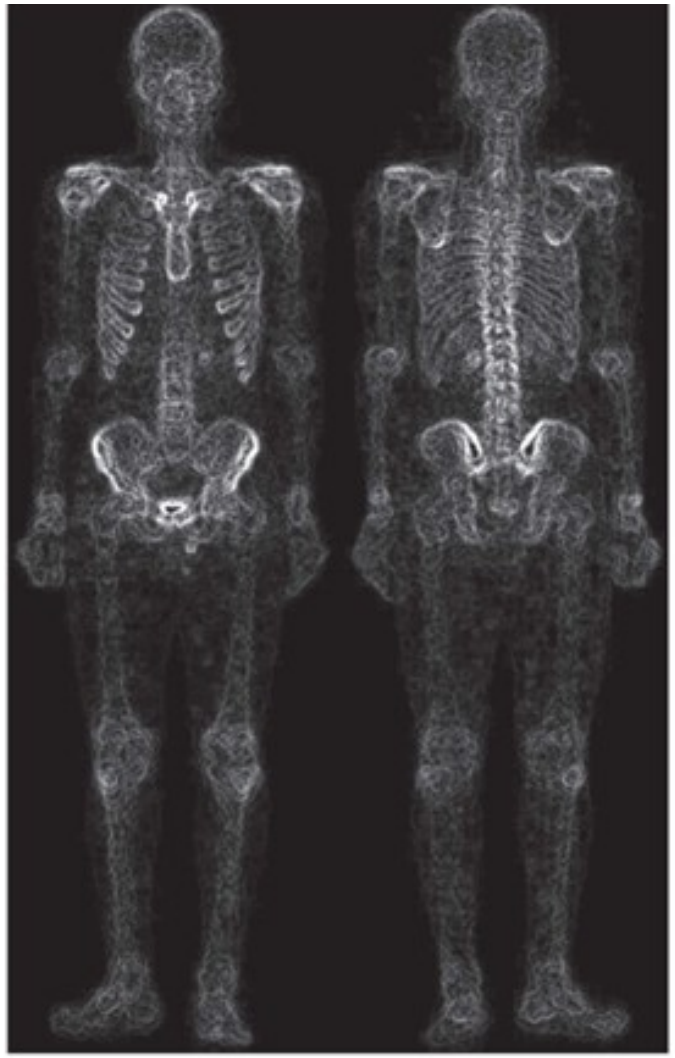


THE SHARPENED IMAGE OF WHOLE BODY BONE SCAN

Sharpened image obtained by adding the input image and the Laplacian image

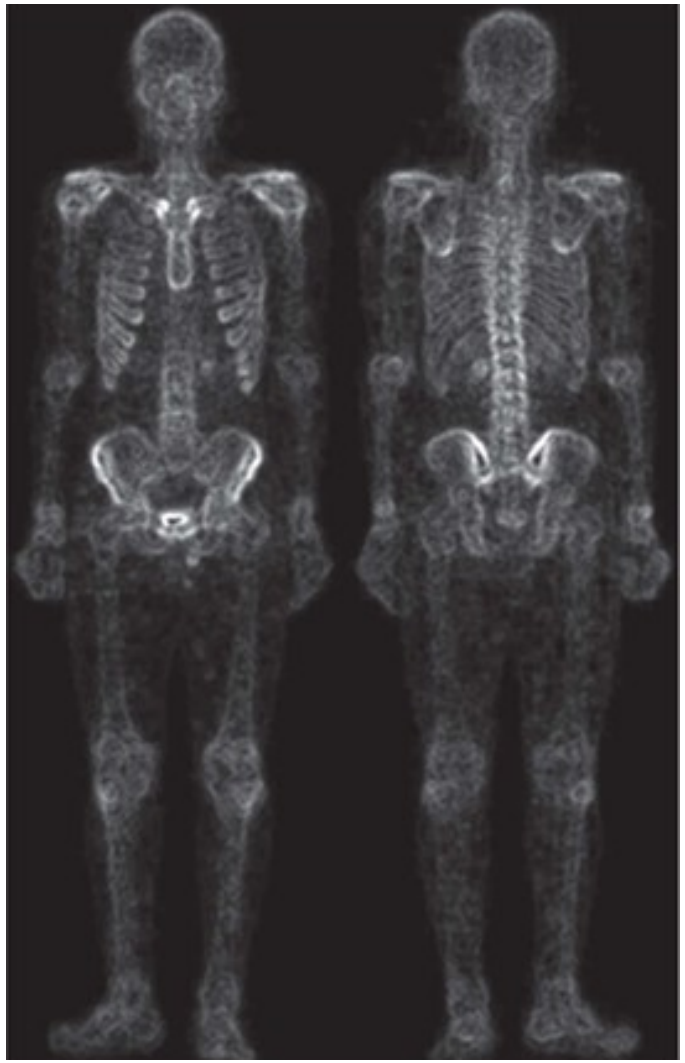


THE SOBEL GRADIENT OF IMAGE OF WHOLE BODY BONE SCAN



THE SMOOTHED SOBEL IMAGE OF WHOLE BODY BONE SCAN

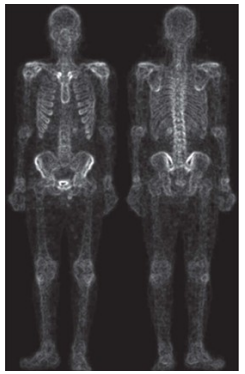
Sobel image smoothed with a 5 x 5 box filter

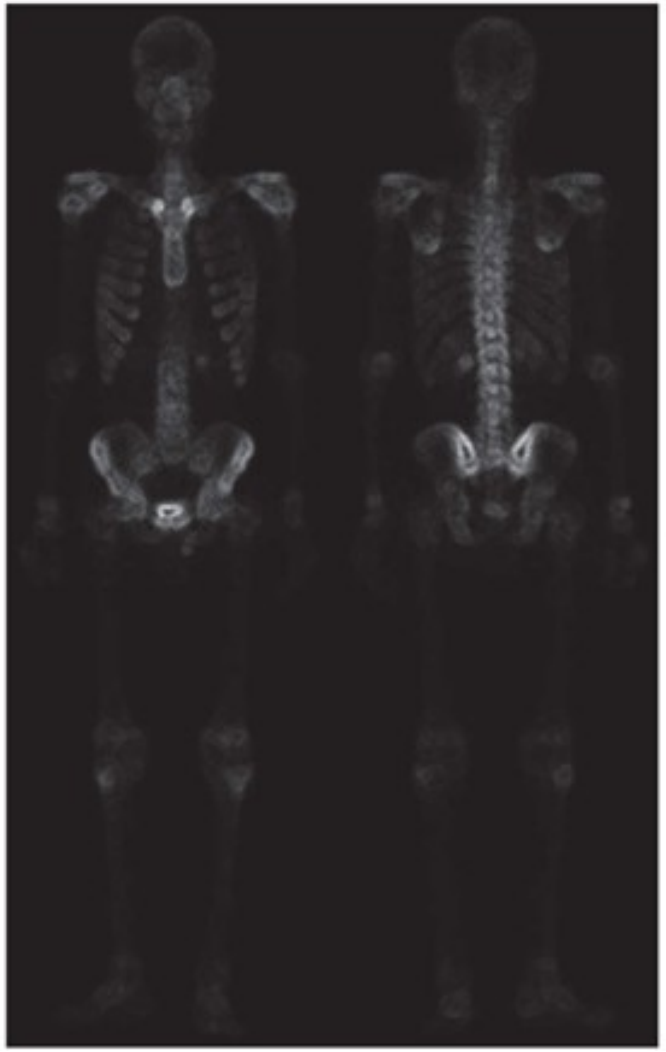


THE MASK IMAGE OF WHOLE BODY BONE SCAN

Mask image formed by the product of the Laplacian image and the smoothed Sobel image

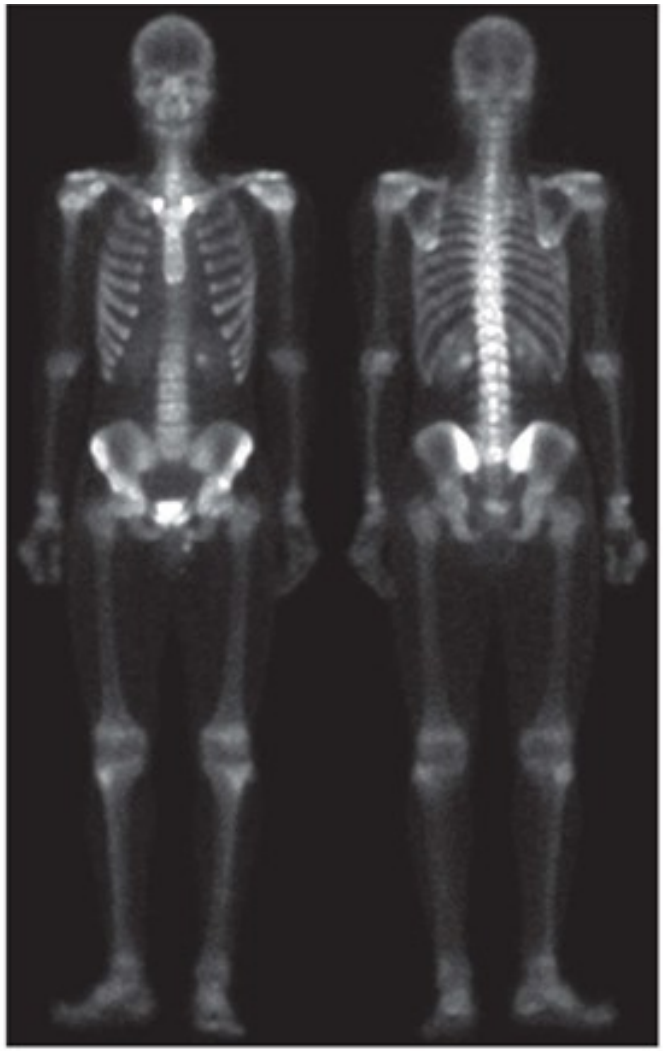






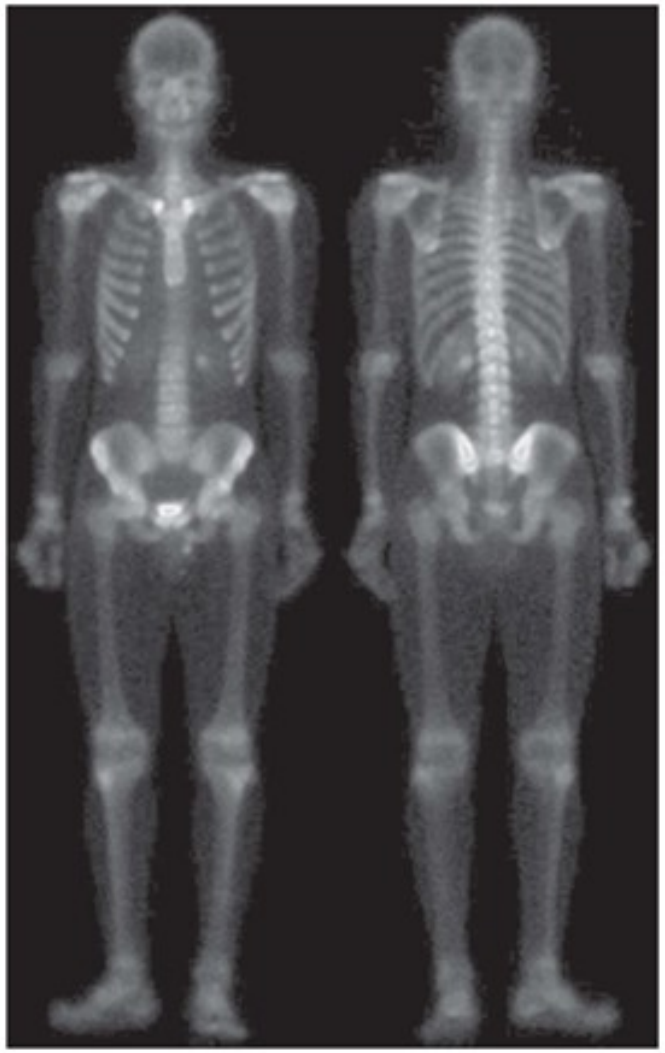
THE SHARPENED IMAGE OF WHOLE BODY BONE SCAN

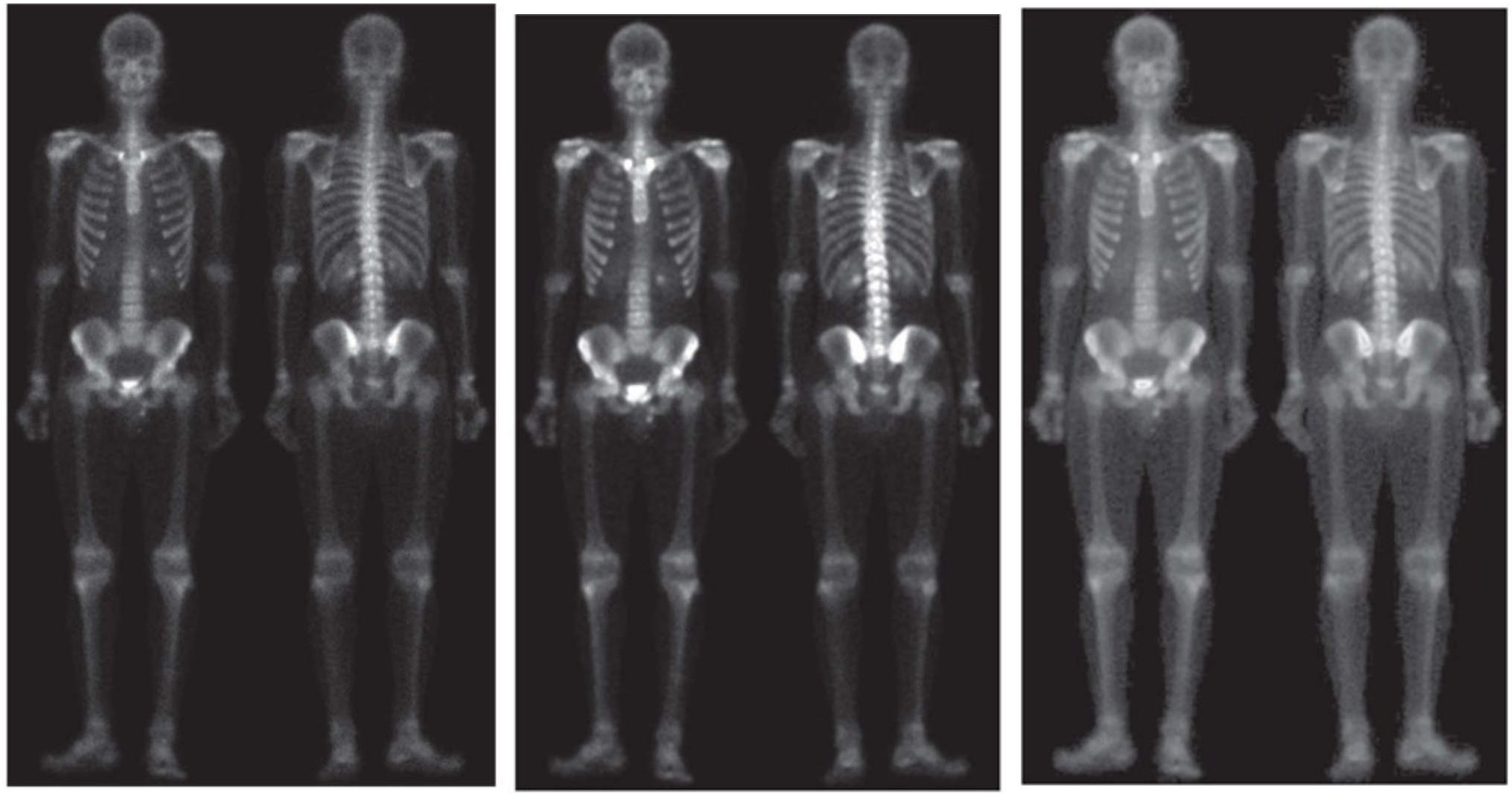
Add the original image with the mask image



THE TRANSFORMED IMAGE OF WHOLE BODY BONE SCAN

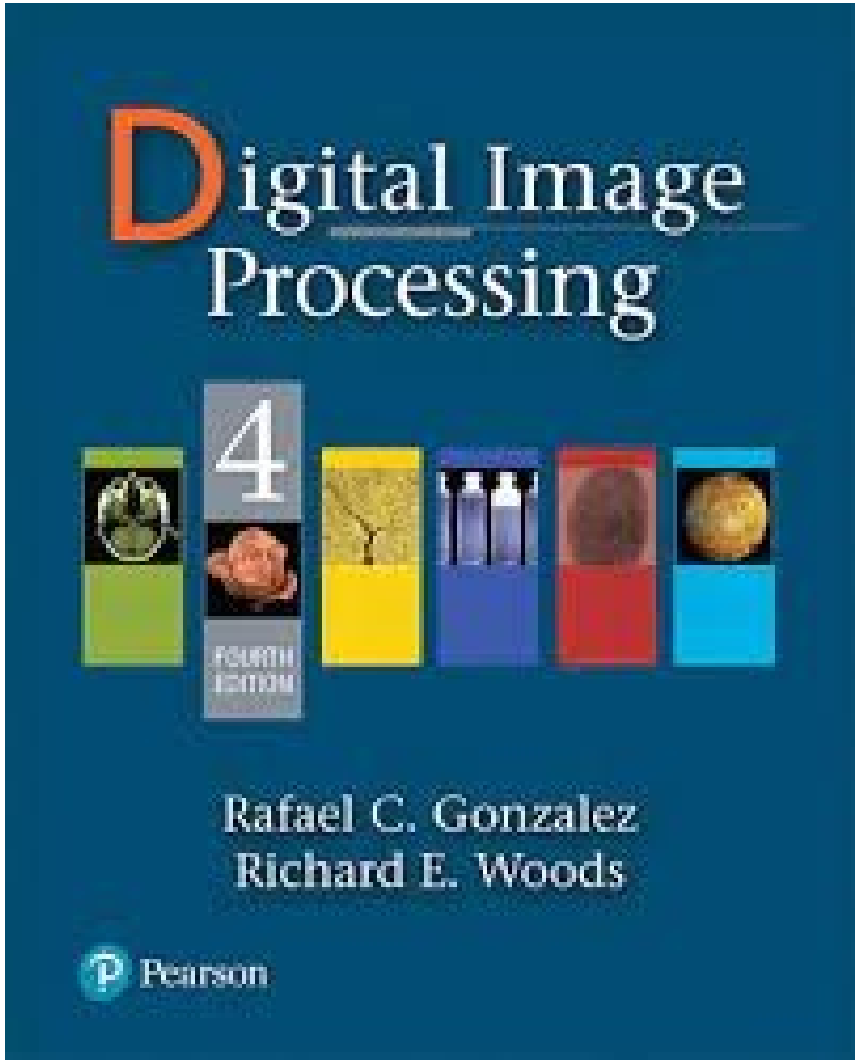
Apply the power-law transformation to the sharpened image ($\gamma = 0.5$)





DAFTAR PUSTAKA

Fundamental steps in digital image processing (Gonzales & Woods, 2018).



TERIMA KASIH